

GenAI - IA-1

Demonstration Screenshots

- Hosted on my pc, using Gratio:
- Homepage:

The screenshot shows a web browser window displaying a Gradio interface for a medical text simplification tool. The title is "Accuracy-Aware Medical Text Simplification Using Generative AI". Below the title, there is a note about the model: "GRU sequence-to-sequence + Luong attention, **proposed model only** (GloVe-50d embedding initialization, fine-tuned, 20 epochs). See the paper and notebook in this repository for the full baseline comparison and evaluation." A warning message states: "⚠ Research prototype, not a medical tool. Trained on a 900-example subsample of the Cochrane medical text simplification corpus (technical review abstracts, not real patient reports) on CPU-only hardware, for a Generative AI course IA. The model frequently invents specific numbers/details not in the source — see the paper's Qualitative & Error Analysis (Section XII) and Ethical Considerations (Section XV). Do not treat its output as medically accurate." The interface has two main text areas: "Technical medical text (source)" and "Simplified output (generated)". The "Technical medical text (source)" area contains a placeholder text: "Paste a technical medical passage here, or click a sample below...". Below this is a "Reference plain-language summary (optional — enables ROUGE / numeric-detail check)" section with a placeholder: "Optional. Leave blank if you don't have a reference." At the bottom of the interface is a large orange button labeled "Simplify". Below the button, there are "Sample test-set examples (real Cochrane pairs used in the paper's evaluation):" with five buttons labeled "Example 1", "Example 2", "Example 3", "Example 4", and "Example 5". At the very bottom, there is a footer with links: "Runs", "Use via API", "Built with Gradio", and "Settings".

- Example 1 (input):

This screenshot shows the same Gradio interface as the previous one, but with the input text filled in. The "Technical medical text (source)" area contains the following text: "Sixteen trials, involving 3361 patients, were included in the review; three of the studies were of magnesium sulphate in addition to nimodipine. Overall, calcium antagonists reduced the risk of poor outcome: the relative risk (RR) was 0.81 (95% confidence interval (CI) 0.72 to 0.92); the corresponding number of patients needed to treat was 19 (95% CI 1 to 51). For oral nimodipine alone the RR was 0.67 (95% CI 0.55 to 0.81), for other calcium antagonists or intravenous administration of nimodipine the results were not statistically significant. Calcium antagonists reduced the occurrence of secondary ischaemia and showed a favourable trend for case fatality. For magnesium in addition to standard treatment with nimodipine, the RR was 0.75 (95% CI 0.57 to 1.00) for a poor outcome and 0.66 (95% CI 0.45 to 0.96) for clinical signs of secondary ischaemia. Calcium antagonists reduce the risk of poor outcome and secondary ischaemia after aneurysmal SAH. The results for 'poor outcome' depend largely on a single large trial of oral nimodipine; the evidence for other calcium antagonists is inconclusive. The evidence for nimodipine is not beyond all doubt, but given the potential benefits and modest risks of this treatment, oral nimodipine is currently indicated in patients with aneurysmal SAH. Intravenous administration of calcium antagonists cannot be recommended for routine practice on the basis of the present evidence. Magnesium sulphate is a promising agent but more evidence is needed before definite conclusions can be drawn." The "Reference plain-language summary (optional — enables ROUGE / numeric-detail check)" section contains the following text: "This review of 16 trials, involving 3361 patients, has found that the outcome after subarachnoid haemorrhage, in terms of survival and being independent in activities of daily living, is improved by treatment with calcium channel blockers (antagonists). If the largest trial is excluded from the analysis, the results are no longer statistically significant, and therefore the evidence is not beyond all doubt. However, given the high likelihood of benefits and the modest risks associated with this treatment, the review authors conclude that calcium antagonists, in the form of oral nimodipine 60 mg every four hours, are useful in patients with subarachnoid haemorrhage from a ruptured aneurysm. Magnesium is another calcium antagonist with promising results, but larger trials with this drug are needed before we can be certain about a beneficial effect." The "Simplified output (generated)" area is currently empty. The "Simplify" button is still present at the bottom. The footer links are also visible.

- (output):

Accuracy-Aware Medical Text Simplification Using Generative AI

GRU sequence-to-sequence + Luong attention, **proposed model only** (Glove-50d embedding initialization, fine-tuned, 20 epochs). See the paper and notebook in this repository for the full baseline comparison and evaluation.

▲ **Research prototype, not a medical tool.** Trained on a 900-example subsample of the Cochrane medical text simplification corpus (technical review abstracts, not real patient reports) on CPU-only hardware, for a Generative AI course 1A. The model frequently invents specific numbers/details not in the source — see the paper's Qualitative & Error Analysis (Section XII) and Ethical Considerations (Section XV). Do not treat its output as medically accurate.

Technical medical text (source)

Sixteen trials, involving 3361 patients, were included in the review; three of the studies were of magnesium sulphate in addition to nimodipine. Overall, calcium antagonists reduced the risk of poor outcome: the relative risk (RR) was 0.81 (95% confidence interval [CI] 0.72 to 0.92); the corresponding number of patients needed to treat was 19 (95% CI 1 to 51). For oral nimodipine alone the RR was 0.67 (95% CI 0.55 to 0.81), for other calcium antagonists or intravenous administration of nimodipine the results were not statistically significant. Calcium antagonists reduced the occurrence of secondary ischaemia and showed a favourable trend for case fatality. For magnesium in addition to standard treatment with nimodipine, the RR was 0.75 (95% CI 0.57 to 1.00) for a poor outcome and 0.86 (95% CI 0.48 to 0.96) for clinical signs of secondary ischaemia. Calcium antagonists reduce the risk of poor outcome and secondary ischaemia after aneurysmal SAH. The results for 'poor outcome' depend largely on a single large trial of oral nimodipine; the evidence for other calcium antagonists is inconclusive. The evidence for nimodipine is not beyond all doubt, but given the potential benefits and modest risks of this treatment, oral nimodipine is currently indicated in patients with aneurysmal SAH. Intravenous administration of calcium antagonists cannot be recommended for routine practice on the basis of the present evidence. Magnesium sulphate is a promising agent but more evidence is needed before definite conclusions can be drawn.

Reference plain-language summary (optional — enables ROUGE / numeric-detail check)

This review of 16 trials, involving 3361 patients, has found that the outcome after subarachnoid haemorrhage, in terms of survival and being independent in activities of daily living, is improved by treatment with calcium channel blockers (antagonists). If the largest trial is excluded from the analysis, the results are no longer statistically significant, and therefore the evidence is not beyond all doubt. However, given the high likelihood of benefits and the modest risks associated with this treatment, the review authors conclude that calcium antagonists, in the form of oral nimodipine 60 mg every four hours, are useful in patients with subarachnoid haemorrhage from a ruptured aneurysm. Magnesium is another calcium antagonist with promising results, but larger trials with this drug are needed before we can be certain about a beneficial effect.

Simplified output (generated)

we searched for studies that compared the effects of a drug - free treatment of a total of patients with copd - we found no small studies that met the inclusion criteria - the studies were conducted in the review - the

Readability (automated formula-based scores — not a human judgment)

	Source	Generated
Flesch Reading Ease	44.9	73.8
Flesch-Kincaid Grade	9.7	6.1

Similarity vs. reference (coarse signal, not factuality)

ROUGE-1	ROUGE-2	ROUGE-L
0.260	0.036	0.130

Numeric-detail preservation heuristic: ▲ no matching number from the reference found in the output (this is the common failure mode documented in the paper's error analysis).

Simplify

Sample test-set examples (real Cochrane pairs used in the paper's evaluation):

Example 1

Example 2

Example 3

Example 4

Example 5

- Example 2: (input and output):

Accuracy-Aware Medical Text Simplification Using Generative AI

GRU sequence-to-sequence + Luong attention, **proposed model only** (Glove-50d embedding initialization, fine-tuned, 20 epochs). See the paper and notebook in this repository for the full baseline comparison and evaluation.

▲ **Research prototype, not a medical tool.** Trained on a 900-example subsample of the Cochrane medical text simplification corpus (technical review abstracts, not real patient reports) on CPU-only hardware, for a Generative AI course 1A. The model frequently invents specific numbers/details not in the source — see the paper's Qualitative & Error Analysis (Section XII) and Ethical Considerations (Section XV). Do not treat its output as medically accurate.

Technical medical text (source)

Eleven studies satisfied inclusion criteria, lasting one week or longer; 949 participants with mostly moderate or severe pain were randomised initially, but fewer completed treatment or had results of treatment. Eight studies were double-blind, two single-blind, and one open-label. None had a placebo only control; eight compared different NSAIDs, three an NSAID with opioid or opioid combination, and one both. None compared an NSAID plus opioid with the same dose of opioid alone. Most studies were at high risk of bias for blinding, incomplete outcome data, or small size; none was unequivocally at low risk of bias. It was not possible to compare NSAIDs as a group with another treatment, or one NSAID with another NSAID. Results for all NSAIDs are reported as a randomised cohort. We judged results for all outcomes as very low-quality evidence. None of the studies reported our primary outcomes of participants with pain reduction of at least 50%, and at least 30%, from baseline; participants with Patient Global Impression of Change (PGIC) of much improved or very much improved (or equivalent wording). With NSAID, initially moderate or severe pain was reduced to no worse than mild pain after one or two weeks in four studies (415 participants in total), with a range of estimates between 26% and 51% in individual studies. Adverse event and withdrawal reporting was inconsistent. Two serious adverse events were reported with NSAIDs, and 22 deaths, but these were not clearly related to any pain treatment. Common adverse events were thirst/dry mouth (15%), loss of appetite (14%), somnolence (11%), and dyspepsia (11%). Withdrawals were common, mostly because of lack of efficacy (24%) or adverse events (5%). There is no high-quality evidence to support or refute the use of NSAIDs alone or in combination with opioids for the three steps of the three-step WHO cancer pain ladder. There is very low-quality evidence that some people with moderate or severe cancer pain can obtain substantial levels of benefit within one or two weeks.

Reference plain-language summary (optional — enables ROUGE / numeric-detail check)

In this review we set out to examine the evidence on how well NSAIDs worked (alone or with morphine-like drugs) in adults with cancer pain. We also wanted to know how many people had side effects, and how severe they were. In April 2017, we found 11 studies with 949 participants. They compared NSAID with NSAID, or NSAID with opioid drug (morphine or codeine). No studies looked at using NSAID together with an opioid-like morphine, which is how they are often used. The studies were small and of poor quality. They used different designs and different ways of showing their pain results. Outcomes important to people with cancer pain were often not reported. Many different NSAIDs were tested, and it was not possible to make sensible comparisons. With an NSAID, initially moderate or severe cancer pain was reduced to no worse than mild pain after one or two weeks in 1 in 4 (26%) to 1 in 2 (51%) people in four studies. Side-effect reporting was poor. Two serious side effects were reported with NSAIDs, and 22 deaths, but these were not related to pain treatment. Common side effects were thirst/dry mouth (1 in 7; 15%), loss of appetite (1 in 7; 14%), sleepiness (1 in 10; 11%), and heartburn (1 in 10; 11%). One in four people stopped taking NSAIDs because the drug did not work, and 1 in 20 stopped because of side effects. The quality of the evidence was very low. Very low-quality evidence means that we are very uncertain about the impact of an NSAID alone for treating cancer pain. We do not know whether using NSAIDs together with an opioid such as codeine or morphine is worthwhile.

Simplified output (generated)

we searched for studies that compared different types of vns - we found no evidence that that endovascular corticosteroids [soris] - and two of these studies were funded to move to be reduced to the risk of developing cvd - we did not

Readability (automated formula-based scores — not a human judgment)

	Source	Generated
Flesch Reading Ease	48.6	66.4
Flesch-Kincaid Grade	11.6	7.3

Similarity vs. reference (coarse signal, not factuality)

ROUGE-1	ROUGE-2	ROUGE-L
0.159	0.025	0.086

Numeric-detail preservation heuristic: ▲ no matching number from the reference found in the output (this is the common failure mode documented in the paper's error analysis).

Simplify

Sample test-set examples (real Cochrane pairs used in the paper's evaluation):

Example 1

Example 2

Example 3

Example 4

Example 5

Runs - Use via API - Built with Gradio - Settings

- Example 2 (without ‘Reference plain language summary):

Accuracy-Aware Medical Text Simplification Using Generative AI

GRU sequence-to-sequence + Luong attention, **proposed model only** (GloVe-50d embedding initialization, fine-tuned, 20 epochs). See the paper and notebook in this repository for the full baseline comparison and evaluation.

⚠ **Research prototype, not a medical tool.** Trained on a 900-example subsample of the Cochrane medical text simplification corpus (technical review abstracts, not real patient reports) on CPU-only hardware, for a Generative AI course IA. The model frequently invents specific numbers/details not in the source — see the paper’s Qualitative & Error Analysis (Section XII) and Ethical Considerations (Section XV). Do not treat its output as medically accurate.

Technical medical text (source)

Eleven studies satisfied inclusion criteria, lasting one week or longer; 949 participants with mostly moderate or severe pain were randomised initially, but fewer completed treatment or had results of treatment. Eight studies were double-blind, two single-blind, and one open-label. None had a placebo only control; eight compared different NSAIDs, three an NSAID with opioid or opioid combination, and one both. None compared an NSAID plus opioid with the same dose of opioid alone. Most studies were at high risk of bias for blinding, incomplete outcome data, or small size; none was unequivocally at low risk of bias. It was not possible to compare NSAIDs as a group with another treatment, or one NSAID with another NSAID. Results for all NSAIDs are reported as a randomised cohort. We judged results for all outcomes as very low-quality evidence. None of the studies reported our primary outcomes of participants with pain reduction of at least 50%, and at least 30%, from baseline; participants with Patient Global Impression of Change (PGIC) of much improved or very much improved (or equivalent wording). With NSAID, initially moderate or severe pain was reduced to no worse than mild pain after one or two weeks in four studies (415 participants in total), with a range of estimates between 26% and 81% in individual studies. Adverse event and withdrawal reporting was inconsistent. Two serious adverse events were reported with NSAIDs, and 22 deaths, but these were not clearly related to any pain treatment. Common adverse events were thirst/dry mouth (15%), loss of appetite (14%), somnolence (11%), and dyspepsia (11%). Withdrawals were common, mostly because of lack of efficacy (24%) or adverse events (5%). There is no high-quality evidence to support or refute the use of NSAIDs alone or in combination with opioids for the three steps of the three-step WHO cancer pain ladder. There is very low-quality evidence that some people with moderate or severe cancer pain can obtain substantial levels of benefit within one or two weeks.

Reference plain-language summary (optional — enables ROUGE / numeric-detail check)

Optional. Leave blank if you don't have a reference.

Simplified output (generated)

we searched for studies that compared different types of vns . we found no evidence that that endovascular corticosteroids (ssris) , and two of these studies were funded to move to be reduced to the risk of developing cvd . we did not

Readability (automated formula-based scores — not a human judgment)

	Source	Generated
Flesch Reading Ease	48.6	66.4
Flesch-Kincaid Grade	11.6	7.3

Simplify

Sample test-set examples (real Cochrane pairs used in the paper's evaluation):

Example 1

Example 2

Example 3

Example 4

Example 5

Runs ⌂ · Use via API 🚀 · Built with Gradio 🍷 · Settings ⚙